

Table 1: 1-NNA-CD & EMD over 100 steps for models trained with different β .

Metrics	Interpolation Coefficient (β)						
	0	0.001	0.01	0.1	0.2	0.5	1.0
1-NNA-CD	0.7289	0.6918	0.6223	0.5968	0.5853	0.5861	0.5989
1-NNA-EMD	0.6647	0.6563	0.6156	0.5884	0.5741	0.5780	0.5869

Table 2: 1-NNA-CD & EMD over 100 steps for models trained using different superset sizes (M).

Metrics	OT Superset Size (M)					
	2048	5000	10000	20000	50000	100000
1-NNA-CD	0.6352	0.5853	0.5853	0.5853	0.5921	0.5725
1-NNA-EMD	0.6254	0.5853	0.5741	0.5627	0.5695	0.5627

Table 1 presents the results. We can observe that directly employing our OT approximation ($\beta = 0.0$) can lead to inferior performance, which can be gradually improved by injecting a small amount of noise into the coupling. Interestingly, the best performance is achieved when compared with other cases at around $\beta = 0.2$. Injecting more noise until arriving at independent coupling does not yield further improvement. These results demonstrate that neither our OT approximation nor independent coupling is optimal in terms of generation quality, and the hybrid coupling is necessary.

OT Supersets Size M . Next, we investigate the importance of constructing sufficiently large supersets for OT computation. Here, we try supersets of varying sizes, starting from 2,048 (number of points to be generated) and progressively increasing the number towards 100,000. As outlined in Section 3.3, we employ the exact OT method for superset sizes of 10,000 or fewer points, and the OT approximation method for larger sets. For each superset size, we also train a flow matching model on the Chair Category and evaluate also its generation quality over 100 inference steps.

Table 2 reports the evaluation results. Overall, we notice that a small superset size usually leads to slightly worse performance, potentially due to overfitting the same target generation points. Increasing the number of points in the superset helps improve the performance. Notably, despite using an approximate OT (introduced in Section 3.3), we still observe some improvement in the generation quality when using a large superset ($M = 100,000$), indicating the benefit of using a large superset outweighs the errors introduced by the approximation.

5 CONCLUSION

In conclusion, we proposed a novel framework, coined as not-so-optimal transport flow matching for 3D point cloud generation. We demonstrated that existing OT approximation methods scale poorly for large point cloud generation and showed that target OT flow models tend to be more complex, and thus, harder to be approximated by neural networks. To address these challenges, our approach introduces an offline superset OT precomputation followed by an efficient online subsampling. To make the target flow models less complex, we proposed hybrid coupling that blends our OT approximation and independent coupling, making our OT intentionally less optimal. We demonstrated that our proposed framework can achieve generation quality on par with existing works given sufficient inference steps, while achieving superior quality with smaller sampling steps. Additionally, we show our approach is effective for conditional generation tasks, such as shape completion, achieving good generation even with five steps. Despite these advancements, there are still several limitations and potential extensions of this work. First, we do not consider other forms of invariance, such as rotational invariance that is present in large shape datasets such as Objaverse (Deitke et al., 2023). Second, we only consider point clouds representing coordinates. It would be interesting to explore the generation of point clouds with additional information, such as points with colors or even Gaussian splitting. Third, we currently focus on generation with a fixed resolution, and it would be interesting to extend our method for resolution-invariant cases, such as those in conditional tasks (Huang et al., 2024). Forth, we assume our dataset does not contain outliers in the point cloud following existing works, and developing a robust learning procedure with a noisy point cloud dataset would be another interesting direction. Fifth, we show empirically that hybrid coupling can reduce the trained models’ complexity (as shown in Figure 5), a theoretical connection between β and the complexities is yet to be studied. Last but not least, we hope that this work can inspire further development around OT maps that are easier to learn and that our proposal, especially for the hybrid coupling, can translate to success at generating other forms of large point clouds such as large molecules or proteins.

6 REPRODUCIBILITY STATEMENT

We demonstrate our efficiency in enabling the reproduction of this work’s results. In particular, we describe the dataset (ShapeNet (Chang et al., 2015)) and the data splits employed in training and evaluation of this work in Section 4. We also cover the evaluation protocol used in this work for assessing performance. Additionally, we include implementation details, such as training specifics, network architecture, and shape normalization for evaluation in Appendix B. Lastly, we outline the procedure of our superset computation and describe the hyperparameter choices in Appendix C.

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